

Lab 27 — NumPy Array Creation and Basic Operations

Aim

To create NumPy arrays using various methods and perform basic mathematical and vectorized operations.

Algorithm / Procedure

1. Import the NumPy library.
2. Create arrays using `array()`, `zeros()`, `ones()`, `arange()`, and `linspace()`.
3. Display the array shape, data type, and dimensions.
4. Perform element-wise addition and multiplication.
5. Calculate the dot product of two arrays.
6. Apply universal functions such as square root and exponential.
7. Display the results.

Program

```
import numpy as np

a=np.array([1,2,3,4])

b=np.zeros((3,3))

c=np.ones((2,4))

d=np.arange(0,10,2)

e=np.linspace(0,1,5)

print("a:",a,"shape:",a.shape,"dtype:",a.dtype,"ndim:",a.ndim)
```

```
print("zeros:\n",b)

print("ones:\n",c)

print("arange:",d)

print("linspace:",e)

x=np.array([1,2,3])

y=np.array([4,5,6])

print("x+y:",x+y,"x*y:",x*y,"dot:",np.dot(x,y))

print("sqrt(x):",np.round(np.sqrt(x),3))

print("exp(x):",np.round(np.exp(x),3))
```

Output

a: [1 2 3 4] shape: (4,) dtype: int64 ndim: 1

zeros:

[[0. 0. 0.]

[0. 0. 0.]

[0. 0. 0.]]

ones:

[[1. 1. 1. 1.]

[1. 1. 1. 1.]]

arange: [0 2 4 6 8]

linspace: [0. 0.25 0.5 0.75 1.]

`x+y: [5 7 9] x*y: [ 4 10 18] dot: 32`

`sqrt(x): [1. 1.414 1.732]`

`exp(x): [ 2.718 7.389 20.086]`

Result

NumPy arrays were successfully created using different methods, and element-wise arithmetic, dot product, and universal functions were successfully performed.